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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations
May 2025

MA1003-1 – DIFFERENTIAL EQUATIONS AND LAPLACE TRANSFORMS
(For AC, EC, EE, VL,)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. The degree and order of the differential equation $1 + \left(\frac{dy}{dx}\right)^3 = c\left(\frac{d^2y}{dx^2}\right)$
A) 1,2
B) 2,1
C) 3,1
D) 1,3
2. In order to find the differential equation of orthogonal trajectories of the curves $f(r, \theta, c) = 0$, which of the following replacements is made
A) $\frac{dr}{d\theta}$ is replaced by $\frac{1}{r} \frac{d\theta}{dr}$
B) $\frac{dr}{d\theta}$ is replaced by $-r^2 \frac{d\theta}{dr}$
C) $\frac{dr}{d\theta}$ is replaced by $-\frac{d\theta}{dr}$
D) $\frac{dr}{d\theta}$ is replaced by $-\frac{1}{r^2} \frac{d\theta}{dr}$
3. Integrating factor of $\frac{dy}{dx} - y = e^x$ is
A) e^x
B) e^{-x}
C) e^y
D) e^{-y}
4. The general solution of the equation $p = \log(px - y)$ is
A) $y = cx + e^x$
B) $y = cx - e^x$
C) $y = c - e^c$
D) $y = cx - e^c$
5. The general solution of $(D^2 + D - 12)y = 0$
A) $y = C_1 e^{-3x} + C_2 e^{-4x}$
B) $y = C_1 e^{3x} + C_2 e^{4x}$
C) $y = C_1 e^{3x} + C_2 e^{-4x}$
D) $y = C_1 e^{-3x} + C_2 e^{4x}$
6. The value of W using the method variation of parameters for the differential equation $(D^2 + 1)y = \tan x$.
A) 1
B) 2
C) e^{4x}
D) e^{2x}
7. The particular integral of $(D - 1)y = e^x$ is
A) e^x
B) xe^x
C) $-xe^x$
D) $x^2 e^x$
8. The complementary function of the differential equation $(D - 4)^2 y = e^{2x}$ is
A) $y = (c_1 + c_2 x)e^{4x}$
B) $y = (c_1 + c_2 x)e^{-4x}$
C) $y = (c_1 + c_2 x)e^{4x}$
D) $y = c_1 e^{4x} + c_2 e^{-4x}$
9. $L(3e^{-2t}) =$ _____
A) $\frac{3}{s-2}$
B) $\frac{1}{s+2}$
C) $\frac{6}{s-2}$
D) $\frac{3}{s+2}$



10. $L(\tan(\frac{\pi}{4})) =$ _____
 A) $\frac{1}{s}$ B) $\frac{1}{s^2}$
 C) $\frac{s}{2}$ D) $\frac{2}{s^2}$
11. If $L\{f(t)\} = \overline{f(s)}$ then $L\{\frac{f(t)}{t}\} =$ _____
 A) $\frac{1}{s} \overline{f(s)}$ B) $\int_s^\infty \overline{f(s)} ds$
 C) $\int_0^\infty \overline{f(s)} ds$ D) $\frac{1}{s} \int_s^\infty \overline{f(s)} ds$
12. $L\{e^{t \cos t}\} =$ _____
 A) $\frac{1}{(s-1)^2+1}$ B) $\frac{s-1}{(s-1)^2+1}$
 C) $\frac{s}{(s-1)^2+1}$ D) $\frac{s+1}{(s+1)^2+1}$
13. $L^{-1}(\frac{1}{(s-2)^2}) =$ _____
 A) e^{2t} B) te^{-2t}
 C) te^{2t} D) e^{-2t}
14. $L^{-1}(\frac{e^{-s}}{s^2}) =$ _____
 A) $u(t-1)$ B) $(t+1)u(t-1)$
 C) $(t+1)u(t+1)$ D) $(t-1)u(t-1)$
15. $L^{-1}(\frac{1}{s^4}) =$ _____
 A) $\frac{t^3}{3}$ B) $\frac{t^3}{6}$
 C) $\frac{t^2}{2}$ D) $\frac{t^4}{6}$
16. $L^{-1}(\frac{1}{s^2+2^2}) =$ _____
 A) $\sin 2t$ B) $\frac{\cos 2t}{2}$
 C) $\frac{\sin 2t}{2}$ D) $\frac{\cos 2t}{t}$
17. The partial differential equation formed by eliminating arbitrary constants from the equation $z = ax + by + ab$
 A) $z = px - qy$ B) $z = px + qy + pq$
 C) $z = px + qy$ D) $z = qx + py + pq$
18. Solution of the partial differential equation $\frac{\partial^2 z}{\partial x^2} = 0$ by direct integration is _____
 A) $z = f_1(y) + xf_2(y)$ B) $z = f_1(y) + f_2(y)$
 C) $z = f_1(y) + xf_2(y) + x^2f_3(y)$ D) $z = x^2f_2(y)$
19. The partial differential equation formed by eliminating arbitrary function from the equation $z = f(x^2 - y^2)$
 A) $py - qx^2 = 0$ B) $py + qy = 0$
 C) $px - qy = 0$ D) $py + qx = 0$
20. One dimensional heat equation is
 A) $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial u}{\partial x}$ B) $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$
 C) $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ D) $\frac{\partial u}{\partial t} = c^2 \frac{\partial u}{\partial x}$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

1. a) Solve $p^3 + 2xp^2 - y^2p^2 - 2xy^2p = 0$
 b) Using the method of Variation of parameters solve the differential equation $(D^2 + 1)y = \sec x \tan x$.
 c) Find the general and singular solution of the equation $p = \log_e(px - y)$.

Marks	BT*	CO*	PO*
6	L*2	1	1
6	L2	2	1
4	L1	1	1

2. a) Solve the differential equation

$$x \frac{d^2 y}{dx^2} - \frac{2y}{x} = x + \frac{1}{x^2}.$$

6 L2 2 1

- b) Solve
- $x \frac{dy}{dx} + y = x^3 y^6$
- .

6 L1 1 1

- c) Solve
- $(D^2 + 3D - 4)y = 12e^{2x}$
- .

4 L1 2 1

3. a) If a body which is originally at
- 80°C
- cools down to
- 60°C
- in 20 minutes, the temperature of air being
- 40°C
- . Find the temperature of the body after 40 minutes from the original.

6 L1 1 1

- b) Solve
- $(D^2 + 4)y = \sin x + \sin 2x$
- .

6 L2 2 1

- c) Find the orthogonal trajectory of the family of curves
- $r = \frac{2a}{1+\cos\theta}$

4 L2 1 1

Unit – II

4. a) If
- $f(t)$
- is a periodic function with period
- T
- then prove that
- $L\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$
- .

6 L1 3 2

- b) Solve
- $x''(t) + 4x'(t) + 4x(t) = 4e^{-2t}$
- ,
- $x(0) = -1$
- ,
- $x'(0) = 4$
- by Laplace transform method.

6 L2 4 1

- c) Find the Laplace transform of
- $e^{2t}(2\sin 4t - 4\cos 4t)$
- .

4 L1 3 1

5. a) Find the inverse Laplace transform of
- $\frac{4s+5}{(s-1)^2(s+2)}$

6 L2 4 1

- b) Rewrite the following function using unit step function and find its Laplace transform:

$$f(t) = \begin{cases} t^2, & 0 < t \leq 3 \\ 4t, & t > 3 \end{cases}$$

6 L2 3 1

- c) Find the inverse Laplace transform of
- $\tan^{-1}\left(\frac{1}{s}\right)$
- .

4 L1 4 1

6. a) Find the Laplace transform of
- $\int_0^t \frac{\cos 3t - \cos 5t}{t} dt$
- .

6 L1 3 1

- b) Using convolution theorem, find the inverse Laplace transform of
- $\frac{1}{s^2(s-1)}$
- .

6 L2 4 1

- c) Find the Laplace transform of
- $t^2 \sin t$
- .

4 L1 3 1

Unit – III

7. a) Derive one dimensional wave equation
- $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$
- .

6 L1 5 2

- b) Form the partial differential equation by eliminating arbitrary functions
- $x + y + z = f(x^2 + y^2 + z^2)$
- .

6 L2 5 1

- c) Solve
- $\frac{\partial^2 z}{\partial x \partial y} = \frac{x}{y} + a$
- by direct integration method.

4 L1 5 1

8. a) Solve
- $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} = 0$
- by the method of separation of variables.

6 L2 5 1

- b) Solve
- $(y+z)p - (z+x)q = x - y$
- .

6 L2 5 1

- c) Solve
- $p(1+q) = qz$
- .

4 L1 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
